EXP NO: 2 Date: 30-7-26	**SUPPORT VECTOR MACHINE (SVM) AND RANDOM FOREST FOR BINARY & MULTICLASS CLASSIFICATION**

AIM:

To build classification models using Support Vector Machines (SVM) and Random Forest, apply them to a dataset, and evaluate the models using performance metrics like accuracy and confusion matrix.

ALGORITHM:

Part A: SVM Model

1. Import necessary libraries
2. Load and explore the dataset
3. Handle missing values if any
4. Encode categorical variables
5. Split dataset into training and testing sets
6. Build SVM classifier using SVC()
7. Train and predict
8. Evaluate the model using accuracy and confusion matrix

Part B: Random Forest Model

1. Initialize Random Forest using RandomForestClassifier()
2. Train and predict
3. Evaluate and compare with SVM

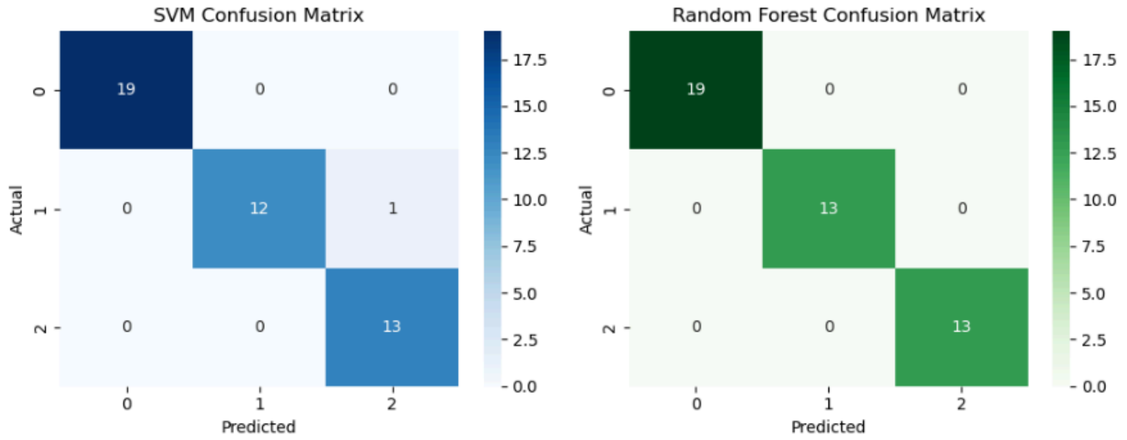
CODE:

```
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

# 2. Load dataset
iris = load_iris()
X = iris.data
y = iris.target

# 3. Feature scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 4. Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.3, random_state=42
)

# ----------------------------
# Part A: SUPPORT VECTOR MACHINE
# ----------------------------

# 5. Initialize and train SVM
svm_model = SVC(kernel='linear')
svm_model.fit(X_train, y_train)

# 6. Predict and evaluate SVM
y_pred_svm = svm_model.predict(X_test)

print("===== SUPPORT VECTOR MACHINE =====")
print("Accuracy:", accuracy_score(y_test, y_pred_svm))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_svm))

# ----------------------------
# Part B: RANDOM FOREST
# ----------------------------

# 7. Initialize and train Random Forest
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

rf_model.fit(X_train, y_train)

# 8. Predict and evaluate Random Forest
y_pred_rf = rf_model.predict(X_test)

print("\n===== RANDOM FOREST =====")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("Confusion Matrix:")
print(confusion_matrix(y_test, y_pred_rf))
```

```python
plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.heatmap(
    confusion_matrix(y_test, y_pred_svm),
    annot=True,
    fmt='d',
    cmap='Blues'
)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.subplot(1,2,2)
sns.heatmap(
    confusion_matrix(y_test, y_pred_rf),
    annot=True,
    fmt='d',
    cmap='Greens'
)
plt.title("Random Forest Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")

plt.tight_layout()
plt.show()
```

OUTPUT:

```
===== SUPPORT VECTOR MACHINE =====
Accuracy: 0.9777777777777777
Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

===== RANDOM FOREST =====
Accuracy: 1.0
Confusion Matrix:
[[19  0  0]
 [ 0 13  0]
 [ 0  0 13]]
```



RESULT:

The Support Vector Machine (SVM) and Random Forest algorithms were successfully implemented for both binary and multiclass classification tasks. The models were trained and tested on the given dataset, and both achieved good accuracy.